

Tracking Vehicles in Cooperative Intelligent Transportation Systems: Attacks, Defense Solutions, and Future Directions

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Abstract—The Cooperative Intelligent Transportation System (C-ITS) is vital in enhancing road safety, improving traffic efficiency, and increasing user comfort for pedestrians and drivers. However, as vehicle communications evolve, security threats that aim to undermine critical security services, such as data confidentiality, integrity, and privacy, have become crucial issues that must be addressed. Privacy, the freedom from interference or intrusion, is also considered one of the most significant challenges in computer networks and connected vehicles. Privacy in C-ITS can be protected by preventing both the collection of personal information and the tracking of vehicles by malicious users. Tracking is often achieved through the periodic transmission of Cooperative Awareness Messages (CAMs), which contain spatio-temporal information such as position and speed. We investigate key tracking-related critical issues in intelligent connected vehicles. We highlight current security challenges and attacks that lead to the unauthorized tracking of connected cars. We discuss various defense solutions proposed in the literature to address these challenges. We classify these solutions into two groups: general (addressing eavesdropping, Sybil, etc.) and pseudonym change (addressing correlation, swap, silent periods, and mix-zones). Finally, we explore robust future strategies to prevent tracking attacks on the C-ITS.

Index Terms—Cooperative intelligent transportation systems (C-ITS), vehicle communication, privacy, tracking.

I. INTRODUCTION

INTELLIGENT Transportation Systems (ITS) have witnessed significant advances in enhancing the safety and comfort of drivers and pedestrians over the last decade. These advances are partly due to the increased cooperation between the devices embedded in the various components

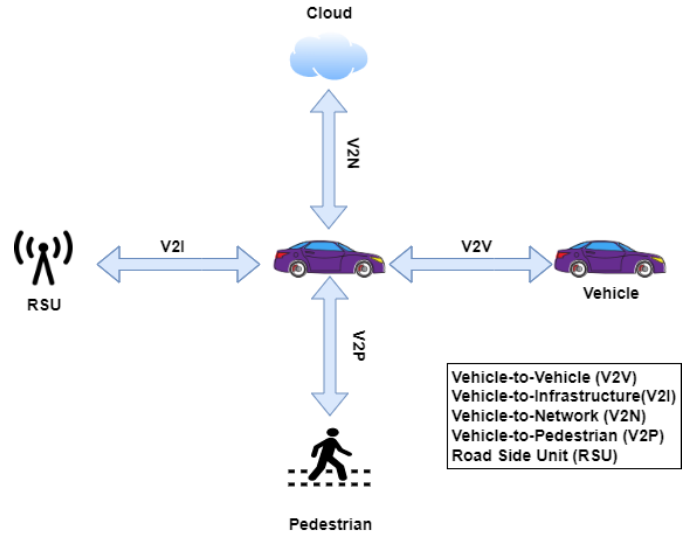


Fig. 1. C-ITS communication types.

(e.g., Road Side Units (RSU), On-Board Units (OBU), and hand-held devices) of the ITS ecosystem which have resulted in C-ITS. This cooperation involves direct Device-to-Device (D2D) communication between all these components, namely, Vehicle-to-Vehicle (V2V), Vehicle-to-Infrastructure (V2I), Vehicle-to-Pedestrian (V2P) and Vehicle-to-Network (V2N) (Fig. 1) [1]. The communication between a vehicle and the other elements of the C-ITS ecosystem is called V2X (Vehicle-to-Everything). The two core technologies behind V2X direct communications are dedicated Short-Range Communications (DSRC) and cellular networks (C-V2X) [2], [3].

From a market perspective, the global market for connected cars is expected to reach 361 billion dollars by 2030 [4]. Automobile manufacturers such as BMW, Toyota, and others spend billions of dollars on software development, connected car technology, and establishing technology centers (for example, the BMW Group Future Mobility Center). Future connected cars will have onboard computers that will allow them to communicate with various entities. This communication will enable multiple functions, such as emergency and information assistance, remote diagnostics, traffic updates, and vehicle-to-vehicle communication. In addition, connected cars will

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allow the creation of intelligent transport systems that can improve traffic flow and reduce road congestion. Mercedes-Benz is an example of a car manufacturer offering its users a range of online services. These services provide drivers with information and assistance, including traffic updates, emergency assistance, and other features [5]. The highly dynamic connections and the time sensitivity that characterize these communications make them attractive targets for attackers. For this reason, companies are working to improve security measures to guarantee better authentication, integrity, and privacy. Furthermore, as vehicles collect more data and become more connected, customers' concerns about trusting vehicle stakeholders with their personal information have increased [6]. This trust is critical for stakeholders. Many important companies (e.g., ESCRYP in Germany) provide robust automotive security systems. However, additional levels of trust must be achieved to establish robust privacy and security rules for customer personal information and tracking. Information privacy, often understood as controlling how personal data is collected, used, and shared [7], is crucial in this context. Personal information includes, but is not limited to, a driver's name, address, email address, location, or any other data that can be linked back to an individual (e.g., credit card numbers, telephone numbers, biometric data). Privacy is a critical concern which must be protected in the environment of connected vehicles. Protecting tracking information is another challenge to preserve consumers' privacy. In V2V communications, vehicles use different techniques to share information and build a cooperative environment between RSUs and vehicles. This information can be used to track users. For example, using identification keys, the Tire Pressure Monitoring System (TPMS) recognizes the car's tires. Each tire is connected to an Electronic Control Unit (ECU) inside the intravehicle network; its ID and pressure are sent to the corresponding ECU every 6 seconds. Tires are kept in cars for extended periods, so TPMS messages are traceable and used to locate the car. In [7], the authors demonstrated that a vehicle can be traced using the IDs of its two tires. Vehicles on the road can be recognized only with prior knowledge of their tire IDs by collecting TPMS status messages.

Most security solutions in C-ITS applications rely on vehicles periodically transmitting unencrypted Cooperative Awareness Messages (CAMs). CAMs frequently save timestamps, IDentity (ID), location, and speed. The use of ID raises the possibility that external eavesdroppers might trace cars by joining consecutive messages with the same ID, revealing the vehicle's whole travel path. To prevent this, researchers proposed providing cars with pseudonyms (instead of real identifiers). Vehicles, however, could still be tracked due to the spatio-temporal information. One solution to overcome this issue is to use many pseudonyms for each car (each of which is only used for a limited and/or context-specific period of time) and switch between them regularly [8], [9]. This does not entirely protect the privacy of tracking information. Many challenges must still be addressed to protect information, such as the widespread transmission of messages (10 messages/sec for traffic safety) containing precise position and time data and the linkability of pseudonymous location

samples with different pseudonyms. If the challenges above are not addressed, an adversary can track vehicles in the long term. If the adversary can cover a vast area and collect a significant number of location samples, the adversary can develop detailed location profiles. Such a scenario is plausible as [10] shows and presents a scenario wherein an adversary collects messages, namely pseudonymous location samples in a particular area. The adversary could link data under several pseudonyms for the specific vehicle. Thus, an attacker can seriously threaten the privacy of the vehicular network user's location. Hence, vehicle tracking cannot be prevented even when pseudonyms are changed [11]. To minimize continuous tracking, some researchers recommended using position privacy techniques such as random silent periods [12] and Mix-Zones [13], [14].

We summarize the contributions of this paper as follows:

- We present a comprehensive survey of vehicle tracking in cooperative intelligent transportation systems.
- We highlight the main tracking issues and describe current tracking attacks and their challenges. To counter these attacks and address the issues highlighted, the paper presents a complete analysis and classification of the available defense mechanisms proposed in the literature.
- Finally, we provide some recommendations and future perspectives for privacy-preserving schemes in C-ITS.

We organize the rest of the paper as follows. Section II presents the related works that have been thoroughly studied and published in the literature. Section III provides some background information on C-ITS with a focus on security requirements. Section IV describes the security tracking attacks faced by connected cars. Section V presents defense solutions to prevent tracking attacks. Finally, section VII makes some concluding remarks. Additionally, Table I presents a list of acronyms used in this paper.

II. RELATED WORKS

The privacy of connected cars has been widely studied in various surveys and publications [15], [16], [17], [18]. These studies cover various aspects of security such as challenges, taxonomies of attacks, and defense mechanisms. Some surveys are solely concerned with privacy [8], [19], [20], [21], while others are concerned with both security and privacy [18], [22], [23]. There is no survey known to date that focuses specifically on vehicle tracking, which is the main contribution of this work. In this paper, we identify critical tracking-related issues, discuss current tracking attacks and challenges, and present a comprehensive analysis and classification of available defense mechanisms. Table II covers current surveys on connected car security issues and their main contributions.

III. COOPERATIVE INTELLIGENT TRANSPORTATION SYSTEMS - BACKGROUND

This section presents an overview of C-ITS from different perspectives: standardization, components, and applications.

A. C-ITS General Overview

1) *C-ITS Communication Patterns*: The communication technologies utilized in C-ITS enable connections between

TABLE I
LIST OF ACRONYMS

Acronym	Description
AI	Artificial Intelligence
BPNN	Back Propagation Neural Network
CAM	Cooperative Awareness Message
CAN	Controller Network Area
CMIX	Cryptographic MIX-zone
C-ITS	Cooperative Intelligent Transportation System
CRL	Certificate Revocation List
CSMA	Carrier Sense Multiple Access
DENM	Decentralized Environmental Notification Message
DSRC	Dedicated Short Range Communications
ECU	Electronic Control Unit
ETSI	European Telecommunications Standards Institute
FL	Federated Learning
GPS	Global Positioning System
IEEE	Institute of Electrical and Electronics Engineers
ITS	Intelligent Transportation System
ML	Machine Learning
MAC	Media Access Control
MARP	MAC layer Attack Resistant Pseudonym
OBU	On Board Unit
PACP	Pseudonymous Authentication with Conditional Privacy
PCS	Pseudonym Change Scheme
PKI	Public Key Infrastructure
RANSAC	RANdom SAMple Consensus
RSU	Road Side Unit
RSSI	Radio Signal Strength Indicator
SAKMP	Secure and Authenticated Key Management Protocol
TAs	Trusted Authorities
TDMA	Time Division Multiple Access
TPMS	Tire Pressure Monitoring System
V2V	Vehicle-to-Vehicle
V2I	Vehicle-to-Infrastructure
V2P	Vehicle-to-Pedestrian
V2X	Vehicle-to-Everything
V2N	Vehicle-to-Network V2N
VPKI	Vehicular Public-Key Infrastructure

various elements, including vehicles, infrastructures, pedestrians, and cloud computing platforms. These C-ITS communications are responsible for establishing connections between vehicles and other entities, such as infrastructure, pedestrians, other vehicles, and networks [15], [25], [26], [27].

- *Vehicle-to-Vehicle (V2V)*: V2V communication allows vehicles with V2V technology to exchange information about other vehicles' speeds and locations over a range comparable to Wi-Fi. This information alerts drivers to potential hazards, reducing accidents and traffic congestion. Within a 300-meter range, V2V communication allows vehicles to detect hazardous traffic conditions, terrain concerns, and weather-related threats.

- *Vehicle-to-Infrastructure (V2I)*: V2I communication facilitates the wireless data exchange between vehicles and roadside infrastructure, such as traffic signals, road sensors, and digital signage. The infrastructure gathers information on traffic conditions, road hazards, and environmental factors, while the V2I system transmits this data to vehicles in real time. This seamless communication improves traffic efficiency, improves safety, and helps autonomous driving. V2I relies on technologies like Dedicated Short-Range Communications (DSRC), cellular networks (e.g., 5G), and Wi-Fi to enable fast and reliable data transmission [28].
- *Vehicle-to-Pedestrians (V2P)*: V2P communication refers to direct connections between a vehicle and one or more nearby pedestrians. This type of communication can also be extended to other potentially dangerous road users, such as cyclists. V2P communication, which can be accomplished directly or via a network infrastructure, allows vehicles and pedestrians to receive warnings of each other's presence. Vehicles can be warned of potentially hazardous road users, such as pedestrians, and pedestrians can be warned of approaching vehicles [29].
- *Vehicle-to-Network (V2N)*: V2N communication (also referred to as Vehicle-to-Cloud (V2C)) can connect vehicles to cloud-based services and data centers, allowing for a wide range of applications. This includes over-the-air software updates for vehicle systems, real-time traffic management, remote diagnostics, and fleet management [30].

2) *C-ITS Components*: C-ITS incorporates two critical components: the OBU and RSU.

- *OBU*: An OBU is a device installed in every vehicle to enable communication with other vehicles and (RSUs). It comprises several components: sensors, storage, and a network device. The sensors (e.g., Global Positioning Sensor (GPS)) gather information transmitted to the OBU. The OBU processes and collects the data contained in messages, which it then wirelessly transmits to neighboring vehicles and RSUs [15], [31].
- *RSU*: An RSU is a stationary device typically positioned along roads or in specific locations, such as intersections or parking lots [32]. It is equipped with a DSRC network device and the ability to communicate with the infrastructure network. The RSU has several key functions, including expanding the communication range of VANETs by transmitting messages to other OBUs and RSUs, executing safety applications such as traffic reports or accident warnings, and connecting OBUs to the Internet.

3) *C-ITS Applications*: Initially, implementing connected vehicles aimed to improve road safety and efficiency by reducing the probability of accidents. These networks serve various functions today and can be classified into three major categories: safety, traffic management, and comfort, as Figure 2 shows. Next, we present a brief overview of each type of application with relevant examples.

- *Safety applications*: The goal of these applications is to decrease the occurrence of accidents and improve the

TABLE II
OVERVIEW OF THE LATEST SURVEY ARTICLES ON CONNECTED CAR SECURITY

Reference	Main contributions
[15]	Examines recent research on QoS, routing, and security in vehicular networks. Next, it contrasts and compares three network simulators: Sumo, MOVE, and VanetMobiSim. It emphasizes that each simulator has its benefits and drawbacks, and the appropriate simulator is chosen based on the simulation's requirements scenario. Finally, it outlines critical challenges, such as scalability, reliability, and security, that must be overcome to adopt VANET technologies successfully.
[18]	Discusses key security concerns for intelligent interconnected cars and presents a thorough overview of the security defenses available. These defenses are divided into four categories: cryptography, software vulnerability detection, network security, and malware detection. The paper emphasizes innovative solutions such as lightweight authentication, software-defined security, and deep learning techniques for improved security protection.
[8]	Presents a comprehensive analysis of pseudonym-based authentication methods in vehicular networks, aiming to balance privacy and security. It categorizes these methods into four types: public key cryptography, identity-based cryptography, group signatures, and symmetric authentication, comparing their advantages and limitations. Additionally, it examines the pseudonym lifecycle, which includes issuance, use, change, resolution, and revocation, across various cryptographic approaches.
[24]	Presents an overview of authentication methods that protect message privacy. These authentication schemes are classified according to the cryptography and the techniques used to protect privacy.
[21]	Presents an overview of essential social theory concepts related to network centrality and analyzes various privacy preservation models for Mobile Social Networks (MSNs) and Vehicular Social Networks (VSNs), focusing on identity, location, and content privacy. This study categorizes privacy threats like identity, location, eavesdropping, manipulation, and service-based attacks while examining countermeasures, including game-theoretic approaches.
[22]	Presents an overview of security, privacy, and trust management challenges in Vehicular Ad Hoc Networks (VANETs), emphasizing their importance for ensuring safe and efficient communications. The paper discusses pseudonym-based authentication techniques and mechanisms for protecting location privacy. It also covers various trust management methods and updates on the latest network simulators, offering insights into current advancements and tools for evaluating and simulating VANETs.
[19]	Presents an analysis and classification of pseudonym change strategies designed to protect location privacy in C-ITS, focusing primarily on techniques that rely solely on these strategies. In contrast, our study takes a broader approach, addressing all aspects of tracking issues, not just those related to pseudonym changes. We identify critical tracking-related challenges, discuss current tracking cyber-attacks, and provide an in-depth analysis and classification of existing cyber-defense mechanisms.
[23]	Analyzes the current automobile security requirements in addition to the architecture and safety components of an open system architecture for the automotive industry.
[20]	Examines various privacy, authentication, and secure message distribution techniques. This paper thoroughly examines the solutions proposed in the last decade to address these concerns. The analysis identifies the constraints and potential challenges that must be addressed in the future.
Our survey	Presents an in-depth survey that focuses on vehicle tracking in Cooperative Intelligent Transport Systems (C-ITS), and discusses the distinct challenges and associated risks of tracking. In contrast to earlier works that mainly discuss general privacy issues, this survey describes tracking attacks and extensively analyzes the various threats aimed at vehicle location data. Additionally, it offers a detailed investigation and classification of the different defense mechanisms proposed in the literature, presenting innovative insights into effective countermeasures for safeguarding vehicles against tracking threats. Finally, this survey provides recommendations and future perspectives for privacy-preserving strategies in C-ITS.

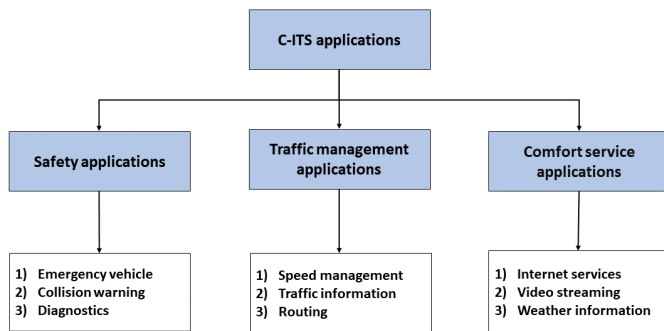


Fig. 2. C-ITS applications.

safety of drivers and passengers. This is achieved by allowing each vehicle to instantly provide a warning when a dangerous event is detected, using techniques such as seat vibrations, tones, visual displays, or a combination of these indicators, as Fig. 2 shows. In an accident, the affected vehicle can continuously transmit critical situation information to nearby cars.

- *Traffic management applications:* C-ITS services include applications other than road safety, such as traffic

management. These applications aim to optimize traffic flow and routes, resulting in shorter travel times, and lower fuel consumption and emissions [33]. Furthermore, these services can help to reduce the number of accidents caused by congested traffic. The benefits of these applications extend beyond road safety, making them a valuable tool for improving transportation efficiency.

- *Comfort service applications:* The primary objective of these applications is to provide comfort and convenience to drivers and passengers. These applications include access to the Internet, multimedia services, messaging, games, and radio channels. This can be achieved through V2I communications, allowing full business services to be accessible from within vehicles [27], [34]. Additionally, V2V communications can facilitate file sharing and video streaming, making extended trips more enjoyable [35].

B. Communication Model and Standards

The cooperation of vehicles from different manufacturers requires standardizing inter-vehicle and intra-vehicle communications. Standardization in C-ITS, as in any other domain,

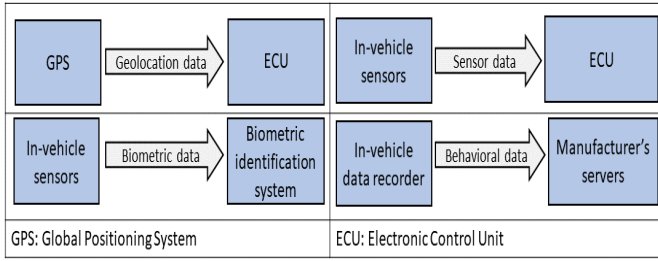


Fig. 3. Data created and stored in cars.

ensures interoperability, supports the corresponding regulations and legislation, and, from a business point of view, expands the related markets [36]. The standardization bodies in Europe, namely, the European Telecommunications Standards Institute (ETSI), and in the US, the Society of Automotive Engineers (SAE), in conjunction with the Institute of Electrical and Electronics Engineers (IEEE) have published several standards covering different aspects of C-ITS which include architecture, messaging, management, access technologies, networking, security, and so on [2] and [37].

From a global perspective, the most widely known V2X-related standardization organizations are from the US (e.g., IEEE and SAE) and EU (ETSI Cooperative-ITS). From a technology perspective, the standards that are commonly used are 802.11-based standards (e.g., IEEE 802.11p [38], [39]) and cellular-based standards (e.g., Cellular-V2X [3], [40]). The IEEE 802.11p standard specifically defines the physical and data link layers for C-ITS (Cooperative Intelligent Transport Systems) nodes, facilitating communication between vehicles and infrastructure. Data can be grouped as follows: location data, sensor data, biometric data, and behavioral data (Fig. 3).

- **Location data:** This data includes longitude, latitude, GPS signal, and GPS time generated by GPS devices embedded in vehicles or by location-based services on mobile phones. While data from GPS devices are stored in the Electronic Control Unit (ECU), data from mobile phones is generally stored locally on the phone or in cloud services. This data gives a complete and detailed record of a driver's movements over time, raising privacy concerns.
- **Sensor data:** Fuel flow, intake air temperature, CO_2 emission, and vehicle Revolutions Per Minute (RPM) are examples of this type of data [41]. ECU collects data from vehicle sensors and stores it.
- **Biometric data:** This is data about the driver's biological features (e.g. face, fingerprint, signature). Biometric identification systems collect and store data from in-vehicle sensors.
- **Behavior data:** This data includes the driver's attention, speed, steering, and braking behaviors along with other diagnostic data used to provide new safety features. It is recorded by the in-car data recorder or stored in the manufacturer's servers.

The various types of information are exchanged through different "standardized" messages (depending on the geographic region we have mentioned previously). In particular, ETSI defines different kinds of application-level messages

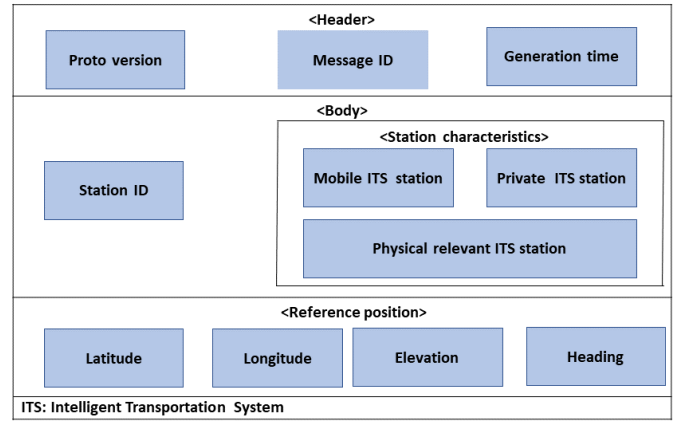


Fig. 4. CAM format.

for road safety and efficiency, namely the CAMs [42] and Decentralized Environmental Notification Messages (DENMs) [43]. DENMs provide early warnings about potentially critical road and traffic events designed to enhance travel safety [44]. Each vehicle transmits CAMs periodically (1-10 times per second). These CAMs provide kinematic data (such as vehicle length and height) and movement information (e.g., position and speed). The type of information CAMs carry and their frequency makes them vital for tracking-related aspects. The CAM includes a header and a body (Fig. 4) [45]. The header contains message-specific information such as message identifier and generation time. The body includes information about the ITS station that sent the message:

- 1) **Station ID:** identifies the sending ITS station.
- 2) **Station characteristics:** indicates the kind of ITS station (mobile, private, and so on).
- 3) **Reference position:** represents the longitude, latitude, and direction of the reference position (e.g., south, north).

The primary use of CAM, as its name indicates, is for cooperative awareness, i.e., to keep vehicles aware of the presence and conditions of other traffic lanes, thus preventing potential accidents [42]. CAMs must be processed by each receiving vehicle, which makes it possible to track the movement and the state of the CAM senders. Many applications rely on vehicle trajectories of successive CAMs to detect traffic patterns.

C. Security in Vehicular Networks

The security of the different exchanges in C-ITS has been addressed by various standardization organizations [46], [47], [48], [49]. The security services that must be ensured in C-ITS for the comfort and safety of passengers and drivers include [20], [50], [51]:

- 1) **Authentication:** is recognized as the primary line of defense against a variety of C-ITS threats [50]. Authentication confirms the identity of communication participants and protects sensitive information by avoiding any unauthorized access [20]. Authentication is a critical feature that must be carefully examined during the design and implementation stages. Key management

and distribution must be accurate and efficient to satisfy authentication requirements.

- 2) *Integrity*: guarantees that the original message is not altered while being transmitted, preventing the unauthorized modification, creation, or deletion of data [52]. It is critical in vehicular networks to verify that the message is not altered throughout its transmission, whether accidentally by degradation factors like fading and noise or intentionally by an attacker. Error detection/correction codes and/or cryptographic primitives must be introduced to achieve this goal. Integrity should also be ensured at the system level, where each device/component must operate as intended. System integrity breaches primarily result from malware injection at various entry points, including V2X communication interfaces, in-vehicle networks (e.g., CAN bus), infotainment systems, and compromised diagnostic tools.
- 3) *Availability*: guarantees that such networks and applications continue functioning in the presence of malicious conditions [53]. Improving the ability to obtain data by/from vehicles is critical in a highly dynamic C-ITS environment; the network should be responsive in real-time. Under typical operating conditions, it is difficult to achieve continuous availability. When a temporary outage occurs through one part of the network, it is critical that connectivity activities continue and that the vehicles remain unaware of any problems. Services must be provided at all times [54].
- 4) *Data confidentiality*: verifies that no unauthorized entities receive the information. Privacy is often considered a sub-category of confidentiality, but given its importance in vehicular networks, a separation is made between data confidentiality and privacy [55]. In other words, data confidentiality can be ensured through traditional encryption algorithms, while privacy requires more complex schemes.
- 5) *Privacy*: Privacy generally includes control over one's information and deciding who can access or process one's data for whatever purpose [56]. In C-ITS, privacy refers to the ability to protect all the personal information related to the driver, vehicle, and passengers, including their identity, location, and so on. We discuss privacy aspects further in section III-D.
- 6) *Non-repudiation*: A vehicle that sent a message should not be able to deny that it sent that message. Such a requirement is essential when an exchange can have financial implications (e.g., requesting a paid service).

Ensuring the security requirements above must address the inherent characteristics of vehicular networks. Designing security solutions for such a constrained context remains a significant challenge. Traditional security algorithms and protocols must be carefully chosen and adapted. Clean-slate approaches might sometimes be needed when existing solutions fail to meet the requirements. The reliability of the messages exchanged in V2X communications is critical, especially when dealing with safety applications. This can be achieved by ensuring the integrity and authenticity of the

messages using asymmetric techniques and digital signature algorithms. Recent advances in lightweight cryptography assist in meeting the real-time constraints of C-ITS. Digital certificates are used, which usually need the deployment of a Public Key Infrastructure (PKI)¹ to manage the different credentials (i.e., certificates) of the C-ITS entities [63].

The traditional PKI does not address privacy because a digital certificate contains the identity of its owner in general, which is not needed to ensure the trustworthiness of a message in V2X communications. It is sufficient to ensure that a "valid" participant has transmitted a message to trust it. A privacy-preserving certificate (used for digital signature) should not include any information that can be used as a link to the identity of a particular vehicle or driver.

D. Privacy in Vehicular Networks

As we previously, privacy is a fundamental security requirement in vehicular networks; any solution, whether security-oriented or not, must be privacy-preserving by design. The privacy requirements in vehicular communication systems include [20], [64]:

- 1) *Minimum disclosure*: Minimize the amount of user information disclosed for basic functionality in vehicular communications.
- 2) *Anonymity*: The message's origin (i.e., the sender) must be kept anonymous among possible senders, also known as the message's anonymity group. However, anonymity is conditional on accountability (i.e., non-repudiation), so trusted authorities should always validate identity resolution.
- 3) *Unlinkability*: It should not be possible to link the relations between items of interest, such as messages, actions, and/or subjects. Any information in vehicular networks becomes privacy-relevant when connected to an identifiable person. Linkability can be direct or transitive (chain of linkable elements).
- 4) *Distributed resolution authority*: In a distributed resolution system, the authorities work collaboratively to establish the link between the anonymous credential and the individual's identity. The process typically involves cryptographic techniques, such as zero-knowledge proofs or secure multi-party computation, to ensure privacy and prevent unauthorized access to sensitive information.
- 5) *Perfect forward secrecy*: Generates new keys for each session so that if one session key is compromised, the confidentiality of other sessions is not affected [65]. In C-ITS, when resolving the credential of an identity, perfect forward secrecy is guaranteed not to expose any information that would compromise the unlinkability of additional credentials of the same user.

The privacy aspects (and consequently the related issues) in C-ITS can be classified into three categories [66]:

¹ Although some PKI-less architectures [57], [58], [59], [60], [61], [62] have been proposed in the literature but standard security architectures are mostly PKI-based [48].

- *Identity privacy*: deals with the privacy of the participant's real-world identity. Participants can be drivers, passengers, pedestrians, and so on.
- *Behavioral privacy*: refers to the privacy of data that describes participants' characteristics (individually or as a group) and actions within the C-ITS, thus protecting them from being profiled.
- *(Geo) Location privacy*: refers to the privacy of location and space. A participant's right is to move within the C-ITS boundaries without worrying about exposing their location information. Tracking a user mainly depends on the privacy of the location information; thus, this category will be the focus for the rest of this paper.

Privacy in C-ITS is achieved through the use of pseudonym certificates.² Pseudonyms are public-key certificates not containing identifying information about a vehicle or user identity. They pseudonymize the vehicle/user to prevent location and identity tracking [69].

1) *Pseudonym Generation Mechanisms*: Several research works [58], [68] have proposed techniques to generate pseudonyms. These mechanisms can be either centralized or distributed.

The authors of [68] proposed a pseudorandom approach where each user is assigned a pseudorandom ID that remains anonymous from all other vehicles. A vehicle creates pseudonyms with the help of an RSU in its neighborhood so that the RSU does not gain any information about the true identity of the node. The authors also proposed a revocation scheme that allows trusted authorities to successfully reveal the identity of a malicious node and revoke it using revocation lists in case of any dispute. The pseudorandom keys are generated using Pseudonymous Authentication with Conditional Privacy (PACP), a new mechanism proposed by the authors. When a user registers a vehicle, the user is issued a ticket with a unique ID used to authenticate the vehicle and generate a pseudorandom key. While this system is useful, it has a large overhead and is vulnerable to identity theft because malicious nodes can copy the ticket obtained from the automobile department.

In [58], the authors proposed a Secure and Authenticated Key Management Protocol (SAKMP). SAKMP is a management technology that distributes public keys based on the user's geographic position. The key is generated using a method that ensures each key is distinct and is based on the position and user's ID. GPS is used to get the position of the users. For the method to create unique keys, a 3D location should be built. The secret is to use the x, y, and z coordinates. This technique offers a secure connection between the RSU and the OBU and it depends on the position and a complex key generation function. Thus, it would be difficult to use in V2V communications. Like all other symmetric encryption-based systems, a large number of keys introduces a significant overhead, particularly for V2V communications. To ensure secure communication in C-ITS, the authors of [62] proposed a pseudonym-based authentication technique. They used group

signatures to create on-the-fly pseudonym keys, allowing users to stay anonymous on the network. In this technique, a user joins a group and is provided with the group's public key, which is used to authenticate their communication quickly. The developers minimized the storage required to develop this technique using group public keys. The usage of group keys, on the other hand, may result in major security breaches. If opponents obtained access to the group keys, they might authenticate their messages.

2) *Pseudonym Change Mechanisms*: Using a single pseudonym is not enough to protect the vehicle/driver's privacy. When a vehicle uses one pseudonym certificate during its lifetime, an adversary who monitors the certificate at various locations can track its location by linking the different messages. In other words, pseudonyms can protect identity privacy but cannot guarantee location privacy.

A key factor in ensuring location privacy is to provide a vehicle with a batch of pseudonyms and to change the pseudonym in use frequently. Vehicles should be able to download a new pseudonyms' set prior to the exhaustion of the current set. The authors of [48] provide more details about the C-ITS security architecture. Most privacy-preserving schemes adopt this approach but differ in the pseudonym change strategy.

In [70], the authors proposed a detailed taxonomy of the different strategies proposed in the literature. Other classifications also exist [69]. Next, we present a simple classification that presents an overview of well-known pseudonym change strategies, which we will address later from an attack and defense perspective.

a) *Parameter-based change*: vehicles can change the pseudonyms based on certain parameters (such as time, number of messages, and traveled distance). A parameter can have a fixed value or a random value (fixed value with a certain random interval around it). Fig. 5 illustrates an example where a pseudonym is changed based on the number of messages signed using that pseudonym. Each vehicle checks if the messages signed with the current pseudonym are above the threshold. If so, the pseudonym and station ID are changed, and messages are sent with the new pseudonym.

b) *Context-based change*: In a context-based change strategy, pseudonyms are altered based on specific conditions, which can be either vehicle-centric or external to the vehicle. Vehicle-centric conditions refer to factors related to the vehicle itself, such as its speed exceeding a certain limit, dropping below a particular threshold, or changing direction. On the other hand, external conditions involve factors (e.g., the density of cars in the vehicle's vicinity, the presence of roadside units (RSUs), and variations in network signal strength) outside the vehicle.

c) *Silent period*: The authors of [71] proposed the concept of the silent period which is the time between using old and new pseudonyms, where a station is not permitted to reveal either the old or new pseudonym. A silent period makes it difficult to determine the time and/or location of a pseudonym change. Since the silence interval affects the temporal connection between two independently received pseudonyms and distorts the place and time where a pseudonym changed,

²Very few research articles have suggested alternatives to pseudonyms [10], [67], [68].

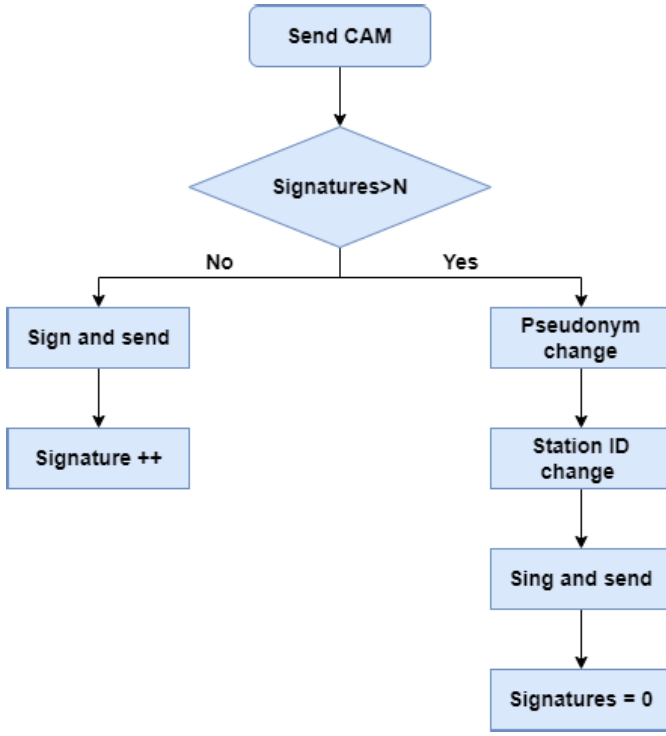


Fig. 5. Counting-based pseudonym change mechanism.

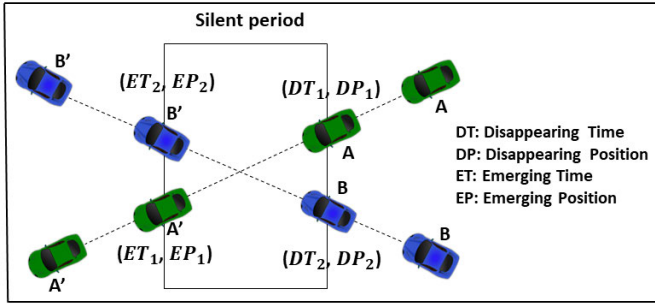


Fig. 6. Silent period pseudonym change.

it becomes difficult to associate two independently received pseudonyms with the same station.

In Fig. 6, car 1 passes from the top to the bottom. In the meantime, car 2 moves from the bottom right corner to the top left. Both cars update their addresses before going silent. The rectangle depicts the silent time. Car 1 reaches the silent period's border with Disappearing Time (DT_1) and Disappearing Position (DP_1), whereas car 2 at DT_2 and DP_2 . Here, we expect that the two cars will reach the border at the same time ($DT_1 = DT_2$). Both nodes stop transmitting packets during the period of silence. After a silent period, cars (1,2) resume frame transmission. The tracking system monitors a novel trace with address A' Emerging Time (ET_1) and Emerging Position (EP_1), as well as another novel trace with address B' Emerging Position (EP_2) and Emerging Time (ET_2). The tracking technique is aware that car 1's address was modified during the silent period. Therefore, because the tracking system identifies two new addresses A' and B' generated after the silent period, it cannot verify if car 1

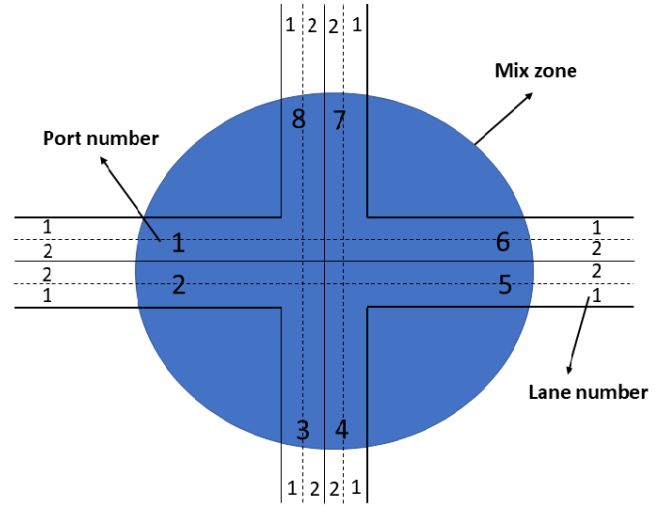


Fig. 7. Mix Zone.

changed its address to A' or B' because car 1 can reach both positions EP_1 and EP_2 from DP_1 due to the silence period introduced during the traveling time.

d) *Mix zones*: the authors of [72] introduced the concept of mix-zone. A mix zone is a geographically specified area in which no location-aware applications are running, i.e., wherein vehicles do not exchange any location-aware message which makes vehicle movements untraceable in this zone. All the vehicles in this area are “mixed” so an attacker cannot determine when the vehicle he/she is tracking left the area. Fig 7 illustrates that the mix-zone is typically placed at intersections or social locations with high traffic and covers multiple entrances and exits. In practice, when approaching a mix-zone, cars would stop transmitting CAMs and keep the radio silent within it. Cars will restart the transmission of CAM messages after exiting the mix zone. Multiple mix-zones approaches [73], [74], [75], [76], [77], [78] have been proposed in the literature and we discuss them in section V.

e) *Pseudonyms swap*: This mechanism swaps certificates between two cars traveling nearby. When a pseudonym certificate is used by a car, this certificate is swapped with another car's certificate. A tracker is confused because the former car will not be the owner of the previous pseudonym certificate. This swap must be recorded on the PKI. This mechanism ensures privacy and introduces another pseudonym certificate-change method. It also reduces the pseudonym certificate costs for drivers. Fig 8 shows that, if the time is greater than the minimum N and the car has a new neighbor, it changes the pseudonym and station ID and sends the CAM.

IV. TRACKING ATTACKS

With a market expected to be worth \$68.0 billion by 2026, [79], intelligent transportation systems are attracting the attackers' attention (regardless of their motives). The research literature [71], [80], [81] identified many potential attacks that, while currently theoretical, still pose a risk. However, real-world attacks have occurred in recent years, as noted in [82]. This section focuses on a specific type of ITS attack that

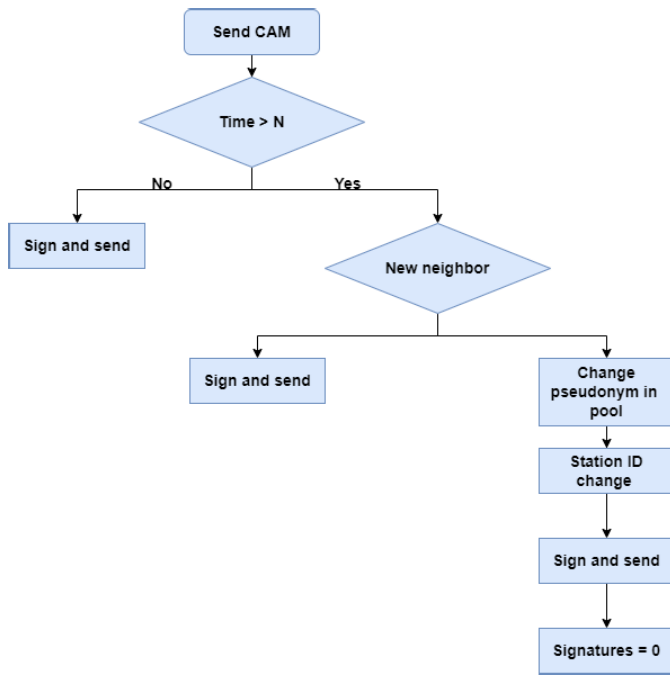


Fig. 8. Pseudonym swap technique.

targets the privacy of ITS users' positions: tracking attacks. Such attacks target the vehicle's security and entail tracking information at different times (generation, processing, transmission) [83]. We can broadly categorize attacks that target tracking into general attacks and attacks against pseudonym change mechanisms (e.g., silent periods, mix-zones).

A. General Tracking Attacks

General tracking attacks are those that, in addition to targeting the privacy of the location data, also allow an attacker to track vehicles. The most widely known attacks in this category are:

1) *Eavesdropping Attacks*: Eavesdropping, in general, aims to steal confidential data. In the context of tracking, it involves monitoring the wireless network, analyzing the packets or messages that are transmitted, and examining their contents to capture the position data and thus be able to track the vehicle's movements [67]. For example, the authors of [11] proposed an anonymous periodic messaging vehicle tracking system in which the attacker model eavesdrops on all network messages. While it may be challenging for an outside attacker to cover the entire network, this model still applies to cases where the attacker has compromised the deployed RSUs.

2) *Sybil Attacks*: They are commonly used in peer-to-peer networks wherein the attacker has greater control over the network due to many "fake" users [84]. In C-ITS, in a Sybil attack, a vehicle pretends to have multiple identities; it uses these different identities while sending messages to other vehicles to change the routing path. The receiving vehicles cannot verify if the received messages are from one vehicle or multiple vehicles [82]. From a tracking perspective, a Sybil attack could be used to direct traffic in a specific direction. By sending multiple messages with different

identities, an attacker can trick other vehicles into believing there is traffic congestion, causing them to reroute to track them [85], [86].

3) *Malware Attacks*: Vehicles in C-ITS are often equipped with infotainment systems that usually come with pre-installed applications but are open for installing third-party applications on handheld devices (e.g., tablets and smartphones). This makes it possible to install malicious software (i.e., malware), which, depending on their type and payload, can pose significant security threats. Moreover, by exploiting certain vulnerabilities in the infotainment system, an attacker can gain access to the Controller Area Network (CAN) bus, which raises safety concerns for drivers and passengers [87]. Such attacks have been demonstrated in production vehicles as in [88]. As in traditional computer networks, some malware can propagate to other "vulnerable" vehicles connected to the infected one. In a tracking attack, malware (e.g., adware, spyware) can be used to steal location information, thereby allowing the attacker to track the victim's vehicle [54].

4) *Correlation Attacks*: A correlation attack is a type of general attack against anonymity services that can be applied to vehicular networks [89]. This attack involves eavesdropping to determine the location of a station by measuring the Radio Signal Strength Indicator (RSSI) of frames received from the station. Frequent MAC address broadcasts give sufficient temporal and spatial tracking precision [80]. The station uses a periodic pseudonym update strategy similar to that defined in the Bluetooth anonymity mode [71]. Bluetooth anonymity mode helps prevent position tracking by frequently changing the active device address. However, given sufficient temporal and spatial accuracy, an attacker could correlate two pseudonyms broadcasted at different times by the same vehicle. As the time between stations changing their pseudonyms may be short, the temporal correlation could be used. If the station continues in the same direction and at the same speed as it did previously, spatial correlation can be used. Correlation attacks get simpler as the distance between vehicles increases, the speed of a vehicle decreases, and the precision of determining its location increases. For instance, the authors of [90] proposed a new MAC layer linking attack that can correlate a vehicle's old and new pseudonyms by studying its transmission properties.

B. Attacks Against Parameter-Based Change Mechanisms

In [10], the authors considered the case of a time-based pseudonym change where vehicles change their pseudonyms every 10 seconds. They proposed a model where an attacker can collect inter-vehicle communication messages, notably pseudonymous position data. They investigated how effectively an attacker can create location profiles and demonstrated that he/she can monitor vehicles and their drivers with near-perfect precision.

In [11], the authors proposed a different tracking technique called Nearest Neighbor Probabilistic Data Association (NNPDA). Even with dense networks and numerous CAMs, it is possible to achieve real-time tracking. The tracking precision achieved raises concerns about location privacy

approaches and pseudonyms change regulations, which have been extensively researched in the literature [68], [70]. It is necessary to secure and protect not only pseudonyms or genuine identities but also vehicle data transmitted via beacons.

The authors of [91] examined a pseudonym-changing technique that was proposed by the European C-ITS platform. By using this technique, they simulated a realistic urban traffic scenario in Luxembourg. In general, it is more difficult to connect pseudonyms with basic traffic information in a real-city traffic scenario. However, the connection of pseudonyms is improved by considering additional static information (such as the vehicle's width and length) from the V2X communication, which makes it possible to track cars and create mobility profiles. Therefore, the number of cars in the area and, in particular, their characteristics, as well as the attacker's ability to observe, have a significant impact on the level of location privacy.

1) *Attacks on Silent Period*: Silent periods have been introduced to increase location privacy and thus the complexity of tracking attacks by adding ambiguity to identifying the time and/or location of a pseudonym modification. Different kinds of attacks have been proposed in the literature [71], [81] on silent-period-based mechanisms.

In [81], the authors presented an attack scenario where the attacker's objective is to estimate the travel distance of the vehicle during the silent period. The expected distance $dist$ is calculated by multiplying the silent period time (t_{sp}) with the street speed limit V_s as Equation (1) shows.

$$dist = t_{sp} * V_s \quad (1)$$

The authors used a modifier named f_d to consider cars that travel slower than anticipated due to heavy traffic. Equation (2) determines the car distance traveled.

$$dact = t_{sp} * V_s * f_d \quad (2)$$

A *Score* is assigned to each potential car by comparing the actual distance $dact$ to the expected distance $dist$ (Equation (3)):

$$Score = |dist - dact| \quad (3)$$

Each vehicle's score is computed and the vehicle with the lowest score is selected. The data used in this attack (speed, position) is included in the CAMs sent.

In [81], the authors presented a more complex attack on the silent period. They developed a sophisticated attack to determine the precise location (x) of a vehicle after a silent period during which vehicles typically do not change direction frequently. To perform this attack, they employed a vector that included the vehicle's direction of travel and the anticipated distance it would cover during the silent period. The effectiveness of the attack is determined by calculating the difference between the estimated position (E_x) and the actual position (l) of the vehicle as represented by Equation (4):

$$Score = |E_x - l| \quad (4)$$

The authors also considered the scenario where vehicles are near the expected point but traveling in the incorrect direction.

These vehicles are excluded by setting a maximum deviation for the direction vector of the vehicles in the form of an angle. The simulation results for different ITS city scenarios (Berlin, New York, and Puebla) showed a higher percentage of 57.87% for the simple attack and 78% for advanced attacks.

2) *Attacks Against Mix Zones*: Pseudonym-change schemes based on mix-zones are also prone to attacks. The authors of [81] presented two classes of attacks targeting such schemes: simple and advanced.

The authors presented a simple attack scenario where an attacker tracks vehicles across a mix-zone by linking entering and exiting vehicles and calculating each entering vehicle's exact exit point and exit time. Several parameters should be taken into consideration to calculate the time expected for the vehicle to exit the mix-zone (Equation (5)):

- 1) The distance traveled is equal to twice the radius R of the mix-zone.
- 2) The street speed limit V_{speed} is the average speed.
- 3) The f_t factor is used to account for the possibility that the attacker's overly optimistic estimate of the time required to pass through the mix-zone is too restricted due to traffic lights or other vehicles' access rights interfering with and slowing vehicles down (Fig 7).

$$t_{expected} = \frac{2R}{V_{speed}} f_t \quad (5)$$

To obtain a more accurate estimate of the target vehicle, we should consider additional information. Vehicle drivers prefer to travel on larger roads at higher speeds. The likelihood of a vehicle downgrading from a larger to a smaller street is lower than in the reverse scenario. Nonetheless, when a vehicle exits the mix-zone onto a street that is smaller than the one it entered, the street size weight W_s , which is a value ranging from 0 to 1, has a negative impact on subsequent calculations $W_s \in [0, 1]$. Another observation is that cars usually drive straight through intersections rather than turning. To accommodate for this tendency, the attacker incorporated another factor, the road turning-weight $W_t \in [0, 1]$. The level of the modifier varies depending on the location of the mix zone and the vehicle trajectory. If the vehicle makes a turn, it is low; if it continues straight through the intersection, it is high. Using these modifiers, a score for each conceivable vehicle is then generated, as Equation (6) shows:

$$Score = (1 - W_s)(1 - W_t)|t_{exit} - t_{enter} - t_{expected}| \quad (6)$$

Ultimately, the car with the lowest computed score is chosen.

In the advanced attack, the entrance and exit points and their related lanes taken by vehicles are numbered in order to create a complex attack, as Fig. 7 shows. A comparison is made between a vehicle that enters the mix-zone at lane a with position x and another vehicle that exits the mix-zone at lane b with position y . An attacker is aware of the typical trip time $t_{xa,yb}$ from lane a of position x to lane b of position y as well as the probability $P_{xa,yb}$ that a vehicle uses to cross this route. The *Score* is computed by Equation (7):

$$Score = (1 - \frac{P_{xa,yb}}{2}) \times (|t_{exit} - t_{enter} - t_{xa,yb}|) \quad (7)$$

The time interval between a vehicle's entry at lane a with position x (at t_{enter}) and its potential exit at lane b with position y (t_{exit}) is compared with the average time t_{xa} , t_{yb} as we have previously outlined in a basic attack scenario. The attacker chooses the vehicle with the smallest computed *Score*.

Finally, a *basic attack* aims to determine the expected travel time based on the radius (R) and street speed (V_{speed}). This means that if vehicles enter through the same entrance, their expected travel time will be identical. In contrast, an *advanced attack* attempts to identify the effect of distinct paths by using the average travel time t_{xa} . Both attacks face difficulty in calculating the estimated travel time if the travel duration of vehicles is excessively scattered or polarized. The simulation results obtained showed a success rate of up to 40% in simple attacks and up to 63% in advanced attacks on mix zones for three different cities (Berlin, New York, and Puebla).

In [92], the authors demonstrated that the estimated travel time in the simple and advanced attacks for tracking is insufficient [81]. As a result, they solved this problem by differentiating the travel time from one vehicle from another by considering vehicle and traffic conditions following parameters such as entrance point, exiting point, and entrance lane, as well as context information to compute the estimated time of the target vehicle through the mixed zone.

- 1) P_{en} : entrance point
- 2) P_{ex} : exit point
- 3) L_{en} : entrance lane
- 4) L_{ex} : exit lane
- 5) V_{ent} : entrance vehicle
- 6) V_{ext} : exit vehicle
- 7) N_{bf} : number of vehicles before entrance
- 8) N_{af} : number of vehicles after exit

The authors of [92] proposed a new vehicle tracking technique based on the Back Propagation Neural Network (BPNN) [93]. Theoretically, under the right conditions of structure and weights, the BPNN technique can estimate any non-linear continuous function. The architecture comprises three layers, the first of which is the input layer which contains various factors (P_{en} , P_{ex} , L_{en} , L_{ex} , V_{ent} , V_{ext} , N_{bf} , N_{af}), the second is the hidden layer (empirical formula), and the third is the output layer (expected time) t_{ex} as Fig 9 shows. The time (t_{ex}) required for a vehicle to cross a mix-zone can be described using Equation 8:

$$t_{ex} = (P_{ent}, P_{ext}, L_{ent}, L_{ext}, V_{ent}, V_{ext}, N_{bf}, N_{af}) \quad (8)$$

For each vehicle that leaves the mix-zone after the target vehicle enters, they considered it as the object vehicle being monitored. This allows them to determine all the necessary input parameters required to train the time prediction model. The output (t_{exnn}) of the BPNN model is the estimated time it takes for the two vehicles to pass each other (when the target vehicle and the object vehicle come into proximity or encounter each other while traveling on the road) as Fig 10 shows. To improve the simple and advanced tracking attack techniques, the terms $t_{expected}$ and $t_{xa,yb}$ are replaced with t_{exnn} resulting in two new equations (Eq. 9 and Eq. 10) for

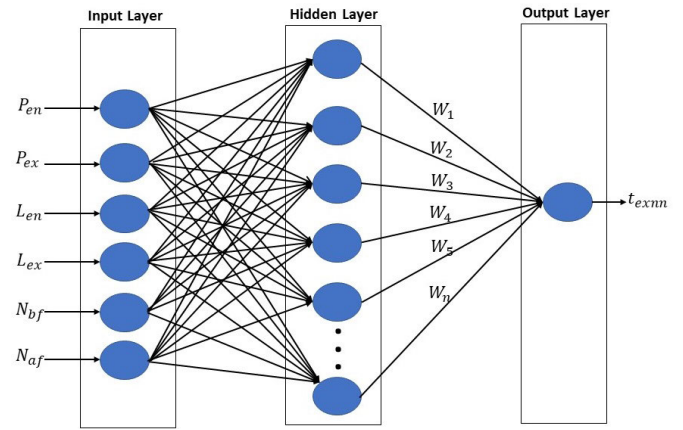


Fig. 9. BP Neural Network.

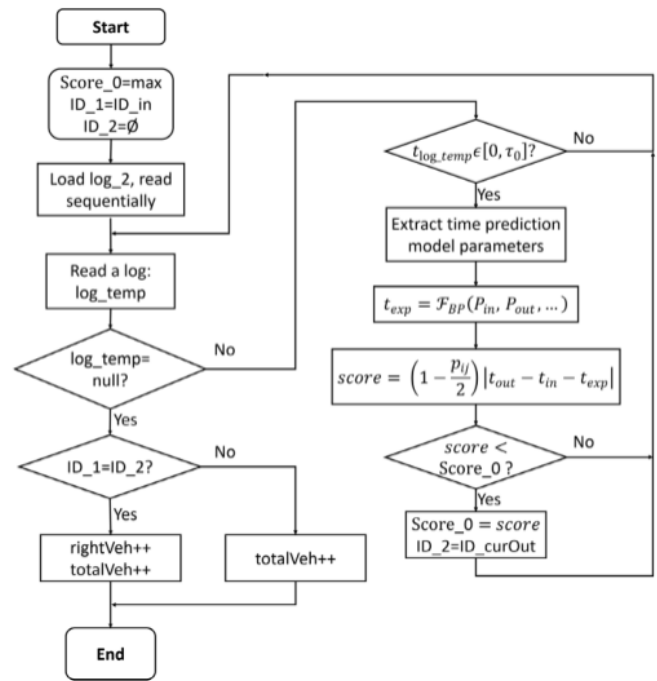


Fig. 10. Tracking Model.

two new attackers: the BP Neural based Simple attacker and the BP Neural based Advanced attacker.

$$Score = (1 - W_s)(1 - W_t)|t_{exit} - t_{enter} - t_{exnn}| \quad (9)$$

$$Score = (1 - \frac{P_{xa,yb}}{2}) \times (|t_{exit} - t_{enter} - t_{exnn}|) \quad (10)$$

When identifying the correct exiting car, only cars that exit the mix-zone after the target car has entered it and within 5 times the average pass time are considered. This is because it is unlikely that any car will remain in the mix-zone for this time. This time-gating approach improves the effectiveness of tracking without significantly lowering the tracking rate. Once the scores have been calculated using equation 9 or equation 10, the car with the smallest score is selected. The real IDs of the resulting enter-exit car pairs are compared to evaluate the attack (Fig. 10). If the IDs of the two cars match, the attack is deemed successful. Formally, the tracking rate is

as follows: Equation (11) shows the tracking rate $Track_{rate}$:

$$Track_{rate} = \frac{\sum_{i=0}^N attack}{N} \quad (11)$$

Equation (12) shows the formula for identifying the attack:

$$Attack = \begin{cases} 0, & \text{if } ID_{in} \neq ID_{out} \\ 1, & \text{if } ID_{in} = ID_{out} \end{cases} \quad (12)$$

The authors of [92] thoroughly examined the limitations in the BPNN-based model and found that its inefficiencies are largely because of many factors used in neural network prediction. To address these issues, they investigated the relationship between these factors and the training aim (i.e., car's pass time), simplifying the factors and customizing the neural network topology (good self-adaptation and its advantage in solving multi-input fitting problems). They demonstrated that their customized time prediction model was more effective and produced superior tracking outcomes (the performance of the model in accurately predicting or estimating the car's pass time). The simulation results obtained showed an average success tracking rate equal to 81.52%.

V. DEFENSE SCHEMES AGAINST TRACKING ATTACKS

This section presents anti-tracking methods for vehicular networks (Table IV). We classify them using the same classification criteria for the attacks we discussed in section IV.

A. Defense Against General Attacks

Vehicular networks can be protected against general attacks used for tracking purposes using the same "general" defense mechanisms. For instance, eavesdropping can be prevented by enforcing the confidentiality of sensitive data through encryption. Sybil attacks can be thwarted by implementing strong authentication mechanisms. In this section, we present defense solutions that are more "customized" for vehicular networks.

1) *Defense Against Eavesdropping Attacks*: Cui et al. [94] proposed a privacy-preserving approach for car location utilizing the concepts of k-anonymity and path confusion. Cars can create dynamic virtual locations in response to their surroundings to deceive information about their driving path, thereby enhancing privacy. They employed the entropy of anonymity set (a measure of uncertainty in identifying a specific car among a collection of vehicles; a higher entropy denotes more confidentiality) and tracking success ratio (a lower tracking success ratio suggests that the privacy-preserving scheme is effective at preventing accurate tracking of vehicles) as criteria to determine the degree of privacy. The results showed that their proposed scheme increased the entropy of the anonymity set by 100% when compared with other schemes while decreasing the tracking ratio to 1%.

The authors of [95] developed a privacy-preserving location strategy that utilizes a filter to differentiate between real and fake messages, incorporating the concept of Chaff messages. The Chaff technique helps protect the content of transmitted messages from eavesdroppers by adding fake data, making it more difficult to trace or link messages to their original

senders. However, the authors noted that this approach could still be susceptible to link attacks, potentially compromising system security. They emphasized that message authentication and sender identity verification improvements are essential for safety applications to address these vulnerabilities effectively.

2) *Defense Against Sybil Attacks*: The authors of [96] proposed a system for detecting and tracking sybil vehicles through a collaborative effort among RSUs. In their approach, three RSUs employ triangulation to determine node positions, which are then shared between RSUs to check for conflicting locations and identify any malicious (i.e., sybil) vehicles. They can detect the trajectory and various identities of sybil vehicles by creating a list of detected vehicles and tracking their pattern of presence in each RSU's coverage zone. Furthermore, RSUs validate messages against sybil nodes and use digital signatures to reduce the impact of false messages in the network.

In [97], the authors proposed an intrusion detection system that uses a signature-based approach and the plausibility concept for detecting fake car movement information. The concept of plausibility [98] involves evaluating the reliability of vehicle movement information based on various factors, such as speed, direction, location, or behavior. By comparing the observed data with the signatures or motion patterns of the original vehicle, the system can determine if the information is likely to be fake or has been manipulated. The intrusion detection system can detect false congestion attacks (sybil attacks) and denial of service attacks, even when a valid movement is used. This technique can identify a single fake car.

In [99], the authors developed a presence evidence system to detect Sybil attacks using position verification. They employed a RANdom SAMple Consensus (RANSAC)-based method [100] to improve the car's location estimation by analyzing the car's signal-strength distribution. RANSAC is an outlier detection method commonly used in computer vision. The presence evidence system determines the estimated location of a claimer car that sends a beacon message at a beacon interval for adjacent discovery.

B. Protecting Pseudonym Change Schemes

The authors of [83] proposed random encryption periods in mix-zone. Every OBU is given a global symmetric key, that it uses to enter an encrypted zone where a certificate update for pseudonym change is executed. To avoid semantic connection, which refers to the ability to link old and new pseudonyms based on identifiable patterns such as vehicle speed, direction, or behavior, this strategy requires cars to change lanes, direction, or speed while altering their pseudonyms, which makes it hard to achieve this scheme in practice. Since the technique permits any OBU to terminate encryption periods, an internal adversary might utilize this to diminish the anonymity set, putting user privacy at risk. Furthermore, the technique adds significant complexity to key management: for each revocation activity, a novel symmetric key must be provided to all legal cars in the network.

1) *Defense Against Correlation Attacks*: The authors of [101] proposed a vehicular PKI-based system that protects

user privacy by avoiding the linking of pseudonyms according to pseudonym timing data. However, a new Media Access Control (MAC) transmission correlated attack (an attack that exploits patterns in MAC layer transmission to correlate transmissions from the same vehicle) [90] can still be launched to compromise location privacy. The authors of [90] proposed a MAC layer Attack Resistant Pseudonym (MARP) technique for beacon broadcasting in vehicular networks. Initially, they identified the new MAC layer context linking attack enabled by inconsistent pseudonym changes and periodic MAC layer operations used in this technique and then examined the attack using both Carrier Sense Multiple Access (CSMA) and Time Division Multiple Access (TDMA) based protocols. With MARP, vehicles change their pseudonyms and time slots by negotiating with their neighbors to build mix zones depending on their current pseudonym age. Vehicles re-arrange their transmission slots at random when changing pseudonyms to remain unlinkable to the adversaries. The time is utilized for communication between vehicles and RSUs or Trusted Authorities (TAs). This communication may involve requesting pseudonyms or receiving the Certificate Revocation List (CRL) from the TA. When a mix-zone is built, vehicles become unidentifiable by cooperatively releasing their transmission slots and old pseudonyms. In this way, the pseudonyms are rendered unlikely.

2) *Collaborative Protection Schemes*: Based on real-time variables such as traffic density, vehicle speed, and infrastructure availability, the author of [102] proposed a new Pseudonym Change Scheme (PCS) that maximizes when, where, and how frequently vehicles change their pseudonyms. The problems associated with existing PCSs include predictable triggers (such as pre-defined time intervals, events that occur at regular intervals, or fixed-distance-based triggers) that jeopardize privacy, reliance on radio silence, and the depletion of pseudonyms. They recommended using traffic patterns and vehicle environments to overcome these challenges, allowing for simultaneous pseudonym changes with unpredictable, dynamic triggers to optimize anonymity. The main contributions of [102] include an on-demand pseudonym generation and authentication mechanism to prevent depletion, a three-tier architecture for coordinating changes within a Road Side Unit (RSU) coverage area, and a context-aware PCS that removes the need for radio silence while avoiding predictable triggers. The authors of [103] proposed two pseudonym exchange algorithms to enhance vehicle privacy by preventing adversaries from linking a vehicle's pseudonyms over time. The Trigger-Based Exchange Algorithm initiates pseudonym exchanges based on specific conditions, such as entering high-traffic areas or detecting tracking attempts. Once initiated, the vehicle sends a request to a RSU, which validates the request and forwards it to the certificate authority for processing. The RSU-Assisted Exchange Algorithm requires vehicles to submit pseudonym exchange requests to the RSU, which authenticates the requests and coordinates with the Certificate authority to enable secure information exchange. Both algorithms sought to make pseudonym exchanges unpredictable, thereby improving privacy while reducing the computational and storage burden on vehicles and the certificate authority. Simulation

results showed that these methods significantly enhance privacy compared to traditional pseudonym-changing schemes. In [104], the authors proposed a pseudonym-based scheme that integrates pseudonym-swapping and pseudonym-changing strategies to enhance location privacy for connected car users. This approach employed two types of pseudonyms: non-swappable and swappable, enabling secure, RSU-independent pseudonym swaps while addressing revocation failures and reputation attacks. During registration, each vehicle receives one non-swappable pseudonym and several swappable ones. Swappable pseudonyms are used for signing and authenticating beacon messages, while non-swappable ones enable secure exchanges without RSUs. After pseudonym swaps, vehicles report their updated vehicle-pseudonym mapping to the certification authority via a secure transport protocol upon connecting to the network. In [105], the authors expanded this work by integrating both pseudonym change and swap strategies, removing the need for preloaded pseudonyms and utilizing reusable long-term certificates. This revision enabled RSU-independent pseudonym swapping while still adhering to VANET security requirements. Like the [104] approach, vehicles are assigned both non-swappable and swappable pseudonyms, using the former for secure exchanges and the latter for signing messages. After each pseudonym swap, vehicles report the updated pseudonyms to the certification authority, enhancing the system's security and privacy.

3) *Protecting Silent Period Based Schemes*: The organized location-independent radio silences, which are related to the notions of random silent periods [71] and Swing & Swap [106], allow vehicles to stop transmitting CAMs for an arbitrary period of time before altering their pseudonyms.

The concept of silent period [71] (that we discussed in section III-D.2 causes vehicles to stop sending any location-related information while changing the pseudonyms to add ambiguity to the identification of the location and time of a pseudonym change). In the initial model for the silent period, the duration of the silent period is fixed for all cars (i.e., the constant silent period). The tracking technique involves comparing the order of the Emerging Times (ET) (the earliest time when the tracking system monitored a specific car with its new pseudonym) of two cars with their new pseudonyms to the Disappearing Times DT (the latest time when the tracking system monitored a specific car with its old pseudonym) of those with their old pseudonyms. The assumption is that if a car disappears with an old pseudonym and re-emerges with a new pseudonym, it is the same car. As Fig 6 shows, the technique described in this figure assumes that the time a car reaches a silent period is a reliable indicator of its ET. However, this assumption may not always hold, especially if the silent period duration is variable. If a car with identity *A* reaches the silent period at an earlier time than another car with identity *B*, the system may infer that the ET of *A* is earlier than that of *B*. However, this inference may not be accurate if the duration of the silent period is not fixed and can vary from car to car. For example, if *A* has a shorter silent period duration than *B*, it may emerge earlier than *B* even if its emerging time is later. To address this issue, the authors of [71] proposed using a variable length

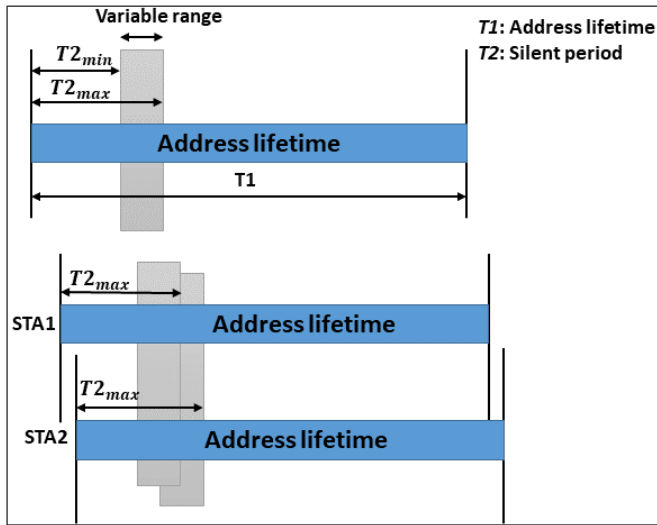


Fig. 11. Variable silent period scheme.

silent period. This approach involves randomizing the length of the silent period for each car, which can help reduce the risk of inferring incorrect ETs based on the timing of the silent period. However, even with a variable duration silent period, there may still be cases where the tracking system cannot link a detected pseudonym to the correct car. For example, if the range of the emerging times for two cars overlaps, the system may not be able to distinguish between them based solely on the timing of their pseudonyms. In such cases, additional information or context may be needed to correctly identify the car associated with a particular pseudonym. The silent period is determined by a random variable varying between $[T2_{min}, T2_{max}]$ as Fig. 11 shows. When the ranges of the emerging times of two cars overlap, issues such as misidentification or ambiguity in associating the pseudonym with the correct vehicle may arise. These challenges could lead to incorrect tracking or difficulty in linking the pseudonym to the right car. In this scenario, the temporal relation between the ET and DT of the two cars may become obscure, making it difficult for the tracking technique to link the old pseudonym of a car to its new one. To address this challenge, the authors recommend that the silent period used for tracking should include both constant and variable periods. The constant period helps to mix the spatial relation between a car's emerging and disappearing points whereas the variable period helps to mix the temporal relation between the car's DT and ET. The use of a constant period helps to mix the spatial relation between the car's emerging and disappearing points by adding noise to the location data. This can help prevent an adversary from accurately identifying the car's path based on the sequence of its pseudonyms. The use of a variable period helps to mix the temporal relation between the car's disappearing time and emerging time by randomizing the duration of the silent period. This can prevent an adversary from accurately inferring the car's emerging time based on the timing of its pseudonym. By including both constant and variable periods in the silent period, the tracking technique makes it more difficult for an adversary to link the old pseudonym of a car to its new one, even in cases where the

TABLE III
NORMAL AND MALICIOUS BEHAVIOR

Car (change flag set to 1)	Silent period	Pseudonym change
Normal	1	1
Malicious	1	0
Malicious	0	1
Malicious	0	0

ranges of their ETs overlap. However, it is important to note that no tracking technique is perfect, and there may still be cases where an adversary can link pseudonyms to individual cars.

The authors of [107] demonstrated how to change a pseudonym successfully and developed an effective pseudonym-changing technique to strengthen users' location privacy in C-ITS. They developed a unique tracking attack known as the cheating attack, which is designed to undermine the performance of the suggested pseudonym change approach. In a cheating attack, the attacker creates CAMs with the change flag set to "1" to trick other cars into thinking it wants to alter its pseudonym when it actually does not. As a result, a car searching for the trigger considers the attacker among the discovered neighbors and changes the pseudonym. Table III demonstrates how normal and malicious cars react to a change in the pseudonym. The normal car enters a silent period and changes its pseudonym, whereas the malicious car (cheater) does the opposite, making it always linkable. This behavior decreases the anonymous set size and makes the linking of pseudonyms easier.

To counteract this threat, they proposed a cheating detection system to identify hostile cheating vehicles.

Some approaches use a silent period jointly with other mechanisms to ensure vehicles' location privacy and thus counter-tracking attacks. Sampigethaya et al. [12] addressed the issue of permitting any vehicle to establish unlinkability between two or more of its locations in the presence of adversary tracking. Unlike prior location privacy approaches in mobile networks, the authors studied the limitations provided by vehicular mobility and vehicular applications within vehicular networks. Specifically, they examined how the movement patterns and operational constraints of vehicles, such as limited control over routing decisions and physical mobility, affect the feasibility and effectiveness of location privacy solutions. They developed a solution that accounts for these limitations and effectively protects location privacy. They concluded that vehicle group navigation can provide location privacy in C-ITS, and to mitigate vehicle tracking, they present CARAVAN. This location privacy approach combines group navigation with a random silent time augmentation technique. The author of [108] proposed a new algorithm for managing pseudonym certificate switching periods between cars to prevent trackers from obtaining drivers' traces. The proposed algorithm executes a short time using a common pseudonym certificate before switching to a new one. During this time, cars sign their messages on the shared pseudonym certificate. The proposed algorithm guarantees that as many vehicles as

possible change their pseudonym certificates during the same slot time, increasing tracker confusion. They assumed that all cars involved acted appropriately, and they did not investigate misbehavior scenarios in the proposed work.

4) *Protecting Mix-Zones Schemes*: One possible real-time location privacy strategy in transportation systems is the mix zone. This method prevents users from tracking attacks by disrupting the continuity of position disclosure and altering pseudonyms. The authors of [76] proposed the MixGroup strategy, which includes social locations and other events where a car can face many other cars. The cars in the Mix-Group communicate with one another to decide if they want to participate in the pseudonym change procedure. In [77], the authors developed an alternative method to construct mix-zone wherein cars build the mix-zone using the radio silence method if their speeds remain under a threshold point for a specific period. Several current techniques [92], [94] study the location and the timing for the pseudonym change. The authors of [78] focused on devising techniques for pseudonym change with dynamically determined mix zones. They used the age of the pseudonym as the most important feature in changing pseudonyms. If the pseudonym age exceeds some threshold, it attempts to change its pseudonym. The authors of [109] proposed a Cryptographic MIX-zone (CMIX) approach for providing location privacy at road intersections. The RSU securely distributes a symmetric key to any incoming car in CMIX to create a mix zone. That symmetric key encrypts all information exchanged in a mix-zone. Furthermore, all cars in the mix-zone must alter their pseudonym certificates. Due to the forced certificate change and the random direction change of every car at crossing points, an adversary on the roadside cannot link a certificate to a specific car thereby providing location privacy. The aggregation of CMIX over the vehicular network creates a mix network, which improves location privacy.

The authors of [110] proposed a Pseudonym Change Protocol (PCP) to enhance vehicle location privacy. This protocol enables vehicles to independently generate legitimate pseudonym certificates without the assistance of Road Side Units (RSUs). Additionally, vehicles can judge the circumstances for changing a pseudonym and get the required information through V2V communication, thereby enhancing the security of the pseudonym change process.

In [108] the authors proposed a new pseudonym-change algorithm. In particular, a vehicle obtains a pool of Pseudonym Certificates (PC) and a Common Pseudonym Certificate (CPC) that is utilized by all cars after it has registered with a Trusted Authority (TA). A car switches its pseudonym certificate when it encounters one or more other vehicles while traveling. The authors of this paper proposed a protocol that allows several cars to switch PCs simultaneously while traveling close to one another; however, there may be some delays. Before upgrading their pseudonyms, all of the nearby cars briefly utilize the CPC. In [95], the authors developed a privacy-preserving location strategy that utilizes a filter to differentiate between real and fake messages, incorporating the concept of Chaff messages. However, the authors noted that this approach could still be susceptible to link attacks. To get complete

unlinkability, the authors of [111] demonstrated the following security and privacy flaws in [95]: attackers may insert fake certificates, affecting the legitimacy of filters and updates, and the insertion may be denied. To solve these challenges, the authors of [111] developed a Vehicular Public-Key Infrastructure (VPKI) approach. First, they used a modified version of the technique devised by [112] to create mutual authentication without disclosing the real identity. Each vehicle's messages are signed with a unique pseudo-identity "certificate" to accomplish this. Second, unlike the [95] technique, they used an adaptive filter to reduce the impact of false positives on the entire filter. Additionally, only RSUs send updates and sign fresh fingerprints to prevent any modifications to the filter. The authors of [113] proposed a privacy security mechanism that relies on altering pseudonyms based on the car's speed and position within or outside established mix-zones. The results obtained demonstrate that the proposed approach is an effective mechanism for protecting car drivers' location information.

VI. FUTURE DIRECTIONS

The future of privacy in connected cars depends on the adoption of multi-layered security techniques that integrate decentralized systems, AI, privacy-preserving communication protocols, and cryptographic innovations. Strong, secure systems that can prevent tracking while maintaining the benefits of connected and autonomous mobility could be developed using emerging technologies like blockchain, Federated Learning (FL), and quantum encryption. Collaboration between technology companies, manufacturers, and government regulators will be crucial to ensuring that these concepts are feasible, scalable, and widely accepted.

A. Federated Learning (FL)

Federated learning is a cutting-edge Machine-Learning (ML) technique that enables linked vehicles to jointly build a common prediction model while maintaining the privacy and decentralization of their data. This technique holds great promise for improving security and privacy in connected automotive systems, especially when it comes to preventing tracking [117], [118]. Federated learning allows models to be trained across multiple vehicles or Intrusion Detection Systems (IDS) without the need to share raw data, thus mitigating the privacy issues associated with conventional centralized learning methods. More precisely, in federated learning, the model is trained locally on each vehicle's data. Only the model updates (called gradients) are sent to a central server. Data is aggregated later to improve a global model, which could be an interesting solution that complies with the General Data Protection Regulation (GDPR) in the European Union. The benefits of Federated Learning for connected vehicles can be summarized as follows:

- Decentralized data processing: Instead of sending data to a central server, federated learning keeps the data on the individual vehicles. This minimizes the possibility that unauthorized parties will intercept or track sensitive data, such as the exact locations or driving patterns.

TABLE IV
OVERVIEW OF SOLUTIONS FOR MITIGATING TRACKING ATTACKS FOR CONNECTED CARS

Category	Attack type	Solution	Performance evaluation
General tracking attack	Eavesdropping	[94] Cars can create dynamic virtual locations to deceive information about their driving path.	Decreases the tracking ratio to 1% and increases the entropy of anonymity set.
	Sybil	[97] Intrusion Detection System (IDS) uses a signature-based approach and applies the concept of plausibility [114][115] to detect fake vehicle movement information.	Increases the detection rate of false congestion attacks (sybil attack).
		[99] Presence Evidence System (PES) to detect Sybil attacks using position verification.	Increases the detection rate of sybil attacks and decreases false positives.
		[96] Three RSUs employ triangulation to determine node positions, which are then shared between RSUs to check for conflicting locations and identify any malicious (i.e., sybil) vehicles.	Reduces the impact of false messages.
Pseudonym change	Correlation	[90] MAC-layer-Attack-Resistant Pseudonym (MARP) technique enables vehicles to adjust their transmission slots and pseudonyms simultaneously, ensuring unlinkability.	Improves the location privacy and the transmission of safety messages.
		[101] Vehicular PKI-based system that protects user privacy by avoiding the linking of pseudonyms according to pseudonym timing data.	Modest computing resources can support a large area with vehicles with low delays and provide privacy protection with a low overhead.
	Silent period	[12] Location privacy approach that combines group navigation with a random silent time augmentation technique.	Enhances the anonymity obtained under correlation tracking with different values for the random silent period during network join.
		[71] The approach randomizes the length of the silent period for each car, which can help to reduce the risk of inferring incorrect emerging time based on the timing of the silent period.	Improves location privacy.
		[108] Novel approach to managing pseudonym certificate switching periods between cars to prevent trackers from obtaining drivers' traces. The proposed algorithm was run for a short period of time on a common Pseudonym Certificate before being switched to a new pseudonym certificate. Cars sign their messages on the same shared pseudonym certificate during this time.	Increases the confusion of trackers.
	Mix-Zone	[109] CMIX approach for providing location privacy at road intersections.	The aggregation of CMIX over the vehicular network creates a mix network which improves location privacy.
		[77] An alternative method to construct a mix-zone wherein cars build the mix-zone using the radio silence method if their speeds remain under a threshold point for a specific period.	High level of location privacy protection without compromising the safety of vehicular networks.
		[111] Vehicular Public-Key Infrastructure-based (VPKI-based) approach to create mutual authentication without disclosing the real identity.	Reduces false positives.
		[95] Privacy location strategy that uses a filter to distinguish between real and fake messages.	Increases the location privacy by up to 76%.
		[116] The dynamic mix-zone approach creates a dynamic mix zone when requested by a car user, regardless of the car's location at the time of the request and enables each car to send safety messages in the mix zone rather than remaining silent.	Improves location privacy.
		[83] Random encryption periods strategy in mix-zone. This strategy requires cars to change lanes, direction, or speed while altering their pseudonyms which makes the scheme hard to deploy in practice. Furthermore, the technique adds significant complexity to key management: for each revocation activity, a novel symmetric key must be provided to all legal cars in the network.	Increases the size of the anonymization set when using a random encryption period.
		[113] Privacy security mechanism based on the car's location and speed to select the car's pseudonym method.	Decreases the continuous tracking percentage and increases tracker confusion, and maximum entropy.

- **Enhanced anonymity:** Federated learning inherently supports data privacy because the individual datasets used for training are never exposed to the central server or other vehicles. This reduces the likelihood of vehicles being tracked through their data.
- **Reduced risk of data breaches:** Federated learning does not require centralized data storage. This reduces the attack surface that hackers can target and minimizes the possibility of significant data breaches that can reveal private data about user behavior and car positions.
- **Deployment of federated learning:** Deploying federated learning in C-ITS with intermittent connectivity requires a robust, flexible approach that guarantees efficiency, privacy, and uninterrupted learning. A hierarchical federated learning framework, where Roadside Units (RSUs) act as intermediary aggregators, allows vehicles to share the local model's updates periodically by transmitting them to a central cloud or Multi-Access Edge Computing (MEC) system when network conditions permit. To address the challenges of unpredictable connectivity, Asynchronous Federated Learning (AFL) [119] enables updates whenever a vehicle establishes a connection. It employs adaptive weighting to prioritize contributions from reliable sources. Privacy protection is enhanced through Differential Privacy (DP), which, rather than adding excessive artificial noise, leverages inherent communication noise. This approach dynamically fine-tunes Differential Privacy (DP) mechanisms to enhance security while preserving model performance. Gradient sparsification, quantization, and secure aggregation further increase efficiency while reducing communication costs and maintaining privacy. To handle data heterogeneity, vehicles also maintain personalized models, while Roadside Units (RSUs) compile information to enhance global updates. This guarantees a learning system that is more robust and flexible. Ultimately, this comprehensive strategy enables the deployment of privacy-preserving, communication-efficient, and robust federated learning in C-ITS, even in environments with variable connectivity. Choosing the right framework is crucial for achieving optimal model outcomes in federated learning. There are numerous tools and platforms for federated learning such as TensorFlow Federated (TFF) [120], NVIDIA FLARE (NVIDIA Federated Learning Application Runtime Environment) [121], FedLab [122], and so on. Each of these frameworks offers unique features that supports different aspects of federated learning, making the selection of the appropriate one vital for successful implementations. However, secure aggregation for federated learning faces significant challenges, including high communication and computational overheads, and vulnerabilities to privacy attacks. To address these issues, various solutions have been proposed in the literature. For instance, [123] proposed SAFElearn, which utilizes Secure Multi-Party Computation (MPC) and Homomorphic Encryption (HE). In [124], the authors proposed Secure Aggregation and Efficient Verification (SAEV) for privacy-preserving federated learning. Additionally, the

authors of [125] developed a novel gradient sparsification framework for secure aggregation called SparseSecAgg. In [126], the authors proposed a novel secure aggregation scheme for Privacy-Preserving Federated Learning (PPFL) in Industry 5.0 ecosystems. It addresses challenges such as heterogeneous data and communication bottlenecks by using a self-attention-based neural network to distill and aggregate data, thereby reducing communication overhead and enhancing efficiency.

B. Quantum Encryption

Quantum encryption presents an innovative way of protecting communications in connected cars, addressing the pressing need for strong privacy protection. Quantum encryption can significantly reduce tracking concerns by providing unbreakable security, guaranteeing anonymity, and promoting collaborative intelligence without sacrificing privacy [127], [128].

- **Secure communication channels:** Quantum encryption enables the safe creation and exchange of cryptographic keys between vehicles and the infrastructure. With the help of quantum key distribution, vehicles may create secure channels for transmitting sensitive data (such as position or navigation information) without running the risk of being intercepted or tracked.
- **Collaborative learning with security:** Quantum encryption can enable collaborative ML among vehicles while ensuring that individual data contributions remain private. Vehicles can securely share insights (e.g., traffic conditions, hazards) without exposing their exact locations or personal information. Future research could explore integrating post-quantum cryptographic protocols with existing vehicular networks to build resilient systems that can withstand not only conventional cyber-attacks but also potential quantum-based threats. This would create safer, more effective, and privacy-preserving vehicle networks by paving the way for a new generation of intelligent transportation systems that are highly collaborative and inherently secure.
- **Integration of Post-Quantum Cryptography (PQC) in connected vehicles ecosystem:** Addressing cybersecurity risks posed by future quantum attacks is essential as we enter the era of quantum computing. The Internet of Vehicles (IoV) must implement cryptographic protocols aligned with the Post-Quantum Cryptography (PQC) standards set by the National Institute of Standards and Technology (NIST) to ensure long-term security [129], [130]. PQC currently relies on lattice-based cryptography, which uses complex mathematical structures formed by combining sets of vectors in large spaces. NIST has recently published encryption and signature standards designed to withstand quantum attacks, including FIPS 203 (Module-Lattice-Based Key-Encapsulation Mechanism for encryption), FIPS 204 (Module-Lattice-Based Digital Signature Algorithm for digital signatures), and FIPS 205 (Stateless Hash-Based Digital Signature Algorithm for hash-based signatures). To strengthen the security of the connected vehicle ecosystem, we must

integrate PQC into security protocols across various components, including On-Board Units (OBUs), Secure Elements (SEs), Electronic Control Units (ECUs), Road-side Units (RSUs), cameras, radar, LiDAR systems, and more.

C. Integration of Blockchain for Decentralized Privacy Management

Blockchain provides a strong foundation for improving privacy in networks of connected cars. Many of the vulnerabilities that permit tracking attacks can be fixed by their decentralized, transparent, and secure nature, enabling intelligent transportation systems that are safer and more reliable [131], [132].

- Balancing audibility and privacy: Blockchain could be employed to create immutable audit trails that record the issuance and usage of vehicle identifiers, helping to ensure transparency and privacy compliance. To prevent setting up a new tracking channel, this must be done carefully. Future research must examine how anonymity might be maintained while preventing unauthorized tracking by utilizing blockchain's transparency.
- Combining mix-zones and pseudonym changes with blockchain: Traditional methods of preventing tracking include mix-zones and changing the pseudonym. By documenting the times and locations of pseudonym changes, blockchain can increase transparency by making sure that these changes are coordinated and unpredictable, which makes tracking more challenging. Future research must examine how blockchain technology might be integrated with current privacy procedures. This would allow for the safe and random changing of pseudonyms without disclosing the true identities of the cars involved.
- Accountability without centralized control: Current centralized systems can suffer from data breaches or misuse by the central authority itself. Blockchain makes it more difficult for any one party to abuse data for tracking by enabling a decentralized method wherein no single entity controls the entire network. Decentralized oversight techniques, in which the network as a whole cooperatively enforces privacy regulations, could be explored and implemented in future systems. Any attempt by an entity to violate these guidelines (for example, by tracking cars without authorization) would be documented on the blockchain, and appropriate remedial measures may be initiated immediately.

VII. CONCLUSION

C-ITS is a critical component in modernizing and improving transportation systems. It leverages advanced communication technologies to enable vehicles, infrastructure, and users to exchange real-time information. However, several essential concerns of security, such as confidentiality, availability, and privacy, must be addressed to ensure safety in C-ITS. This paper presents an in-depth survey of recent research on privacy, particularly regarding vehicle tracking, highlighting current threats and challenges. We analyzed and classified existing tracking defense solutions that have been discussed

and proposed in the current literature. Additionally, we recommended future directions to mitigate tracking attacks. Future research on connected vehicles should adopt multi-layered security strategies integrating decentralized systems, artificial intelligence, and cryptographic innovations. Emerging technologies such as blockchain, Federated Learning (FL), and quantum encryption can enhance privacy protection while maintaining the benefits of connected mobility. We hope this survey provides valuable insights for researchers working on privacy and vehicle tracking within C-ITS.

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